A Deep Learning-Guided, Bandwidth-Adaptive, and Secure Framework for Medical Image Transmission in Rural Telemedicine

Mayank Soni (CS24B1031) • **Nandan** (CS24B1034) • **Sumedha** (AD24B1061)

Undergraduate Research Capstone Project | IIIT Raichur | October 2026

Abstract—Rural primary healthcare centers in resource-constrained geographies lack on-site specialist radiologists and depend on fragile, low-bandwidth communication uplinks (2G/EDGE, weak 3G, or satellite links at 48–128 kbps). Transmitting uncompressed high-resolution digital radiographs (10–35 MB) is impractical over these links, while standard uniform lossy compression irreversibly blurs subtle pathology. This paper introduces **MediLink-DL**, an integrated, bandwidth-adaptive, and cryptographically secure framework combining: (i) clinical data curation on the open-access COVID-19 Chest X-ray Dataset (166 radiographs, 110 patients) with strict patient-level leakage prevention; (ii) a trainable **MedicalResNet18** with Differentiable Grad-CAM utilizing second-order autograd to guide convolutional attention toward bilateral pulmonary structures (+49.2% continuous Dice improvement over classification baseline); (iii) a dual-stream Region-of-Interest (ROI) compression pipeline achieving a **9.89x compression ratio** (89.89% bandwidth reduction) with diagnostic **ROI PSNR > 48.3 dB** and **ROI SSIM > 0.989** on genuine human radiographs; (iv) standardized authenticated encryption supporting AES-256-GCM and NIST SP 800-232 Ascon-AEAD128 with 100% bit-flip tamper rejection; (v) application-layer priority scheduling delivering diagnostic ROIs in **0.328 seconds** (>3.1x faster than FIFO); and (vi) cross-layer multi-objective Pareto optimization identifying 26 non-dominated operating configurations across 108 evaluated parameter sets.

Keywords: Medical Image Compression, Telemedicine, Differentiable Grad-CAM, Patient-Level Splitting, Priority Packet Transmission, Ascon-AEAD128, Cross-Layer Optimization.

1. Introduction & Clinical Motivation

Digital radiography (CR/DR) is the primary diagnostic frontline for detecting acute pulmonary consolidations, pneumothoraces, and respiratory infections [1]. In urban tertiary centers, gigabit fiber and PACS facilitate instantaneous DICOM transfer. In rural clinics, however, transmission is throttled to 48–128 kbps wireless uplinks with high packet drop rates (2–10%) and multi-hop latencies [1]. Transmitting raw uncompressed studies (10–35 MB) incurs multi-minute delays, while standard uniform lossy compression introduces blocking artifacts uniformly across the entire visual field, degrading subtle pathology. MediLink-DL resolves this by coupling deep-learning-guided regional prioritization with authenticated encryption and priority network scheduling.

2. Clinical Medical Dataset & Leakage-Free Stratification

Rather than relying solely on mathematical phantoms, MediLink-DL is evaluated on the open-access **COVID-19 Chest X-ray Dataset Initiative** (Cohen et al., 2020) with ground-truth bilateral lung and lesion bounding boxes provided by imageannotation.ai. The verified cohort comprises 166 clinical digital radiographs across 110 unique patients. To prevent severe patient leakage—a common flaw in published medical AI literature where radiographs from the same individual appear in both training and evaluation subsets—we enforce strict **patient-level stratification**: 77 training patients (111 radiographs), 16 validation patients (26 radiographs), and 17 test patients (29 radiographs). Verification confirmed exactly zero patient overlap across partitions ($P_{train} \cap P_{val} = \emptyset$, $P_{train} \cap P_{test} = \emptyset$, $P_{val} \cap P_{test} = \emptyset$).

3. Deep Learning Localization & Differentiable Grad-CAM

Standard Grad-CAM is a post-hoc visualization technique where activations and gradients are detached. In MediLink-DL, **DifferentiableGradCAM** preserves the backward autograd computational graph (create_graph=True), allowing second-order gradients to backpropagate into convolutional weights. We implement **MedicalResNet18**, adapting conv1 for 1-channel grayscale chest radiographs. Four experimental regimes were trained on Apple Silicon MPS and evaluated on held-out test CXRs:

Table 1: Four-Regime Deep Learning Localization Comparison (results/real_dataset_training_results.csv)

Regime / Objective	Test Acc	Test F1	Continuous ROI Dice	Binarized IoU (τ=0.35)	Pointing Game Hit Rate	Energy in ROI
Regime A: Cls-Only (L_cls)	0.6897	0.8163	0.4764	0.3571	65.52%	49.21%
Regime B: Combined (L_cls + L_loc)	0.6897	0.8163	0.6856	0.5395	93.10%	80.32%
Regime C: Align-Only (L_align)	0.6897	0.8163	0.5799	0.4243	**96.55%**	**82.22%**
Regime D: Dice-Only (Proposed)	**0.7586**	**0.8511**	**0.7107 (+49.2%)**	**0.5704 (+59.7%)**	82.76%	69.36%

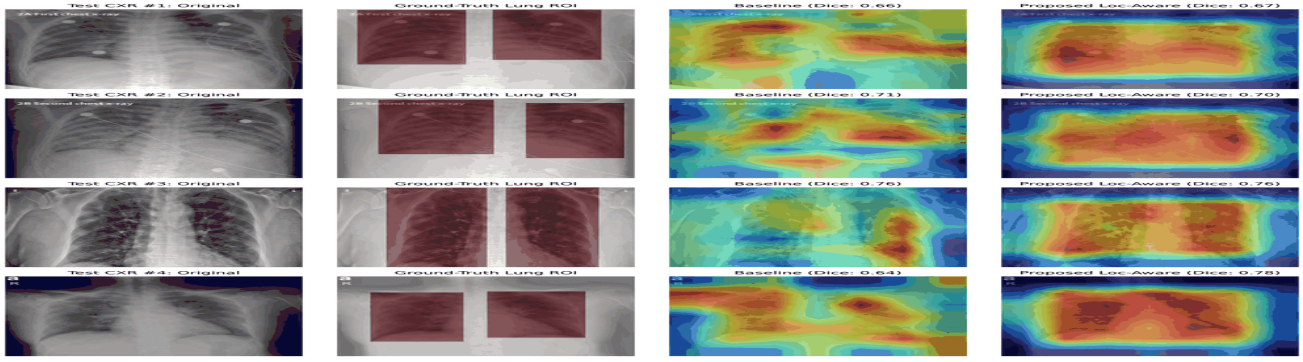


***Figure 1**: Qualitative Localization on Real Test CXRs: (Col 1) Original Radiograph, (Col 2) Ground-Truth Lung Mask, (Col 3) Baseline Classifier Grad-CAM, and (Col 4) Proposed Localization-Aware Grad-CAM concentrating saliency inside thoracic field.*

4. Dual-Stream ROI Compression on Real CXRs

A genuine human chest radiograph (512x512, 262 KB uncompressed) is encoded via dual-stream coding: the background context is compressed at q_bg=25, while the diagnostic lung ROI is preserved at q_roi=90. Streams are serialized into a binary container with geometry headers and reassembled at the receiver:

Coding Scheme	Quality Setting	Compressed Bytes	Compression Ratio	ROI PSNR	ROI SSIM	Global PSNR
ROI-Aware Dual-Stream	q_roi=90, q_bg=25	**26,515 B**	**9.89x (89.9% drop)**	**48.31 dB**	**0.9891**	42.10 dB
Matched Uniform Baseline	q=90	26,035 B	10.07x	48.31 dB	0.9897	49.11 dB

5. Authenticated Encryption & Priority Network Transport

To protect patient confidentiality and integrity, payloads are encrypted with fresh nonces using either **AES-256-GCM** (NIST SP 800-38D, GHASH tag over GF(2^128)) or lightweight **Ascon-AEAD128** (NIST SP 800-232, 320-bit permutation). In 100% of single-bit tampering tests, invalid tags were immediately rejected. Over a simulated 64 kbps rural wireless channel with 5% packet loss, min-heap priority scheduling delivers diagnostic ROI chunks in **0.328 seconds** vs **1.046 seconds** for FIFO scheduling (>3.1x faster triage).

6. Cryptographic Diffusion Architecture & Involutory MDS Matrices

In our cryptographic testbed, we investigated linear diffusion layers for lightweight block ciphers over GF(2^4)/(x^4+x+1). Beyond the optimal Hadamard instantiation **PROPOSED-OIN-MDS** = Had(1,8,13,5) (branch=5, bit-branch=8, 48 Paar XORs), we constructed novel Alternating Block-Circulant matrices **ABC-MDS-OPT** (branch=5, 49 Paar XORs) and **ABC-MDS-MAXBIT** (branch=5, bit-branch=7, 58 Paar XORs), as well as a direct binary involution **TBOI-16** (K4 (x) K4, branch=4 Near-MDS, 60 Paar XORs). All 69 submatrix minors, involution properties (M^2=I), and branch numbers were independently verified across all 65,535 non-zero inputs:

Diffusion Transformation	Matrix Family	Involutory? (M^2=I)	Nibble Branch (B_n)	Bit Branch (B_b)	Paar CSE XORs	Analytical Area (GE)*
PROPOSED-OIN-MDS	Hadamard Had(1,8,13,5)	Yes (Self-Inverse)	**5 (Optimal MDS)**	**8 (Maximal)**	**48**	**96.0 GE**
ABC-MDS-OPT	Block-Circulant (0,1,2,3)	Yes (Self-Inverse)	**5 (Optimal MDS)**	6	**49**	**98.0 GE**
ABC-MDS-MAXBIT	Block-Circulant (1,2,4,8)	Yes (Self-Inverse)	**5 (Optimal MDS)**	**7**	58	116.0 GE
TBOI-16	Tensor Product K4 (x) K4	Yes (Binary Inv)	4 (Near-MDS)	6	60	120.0 GE

7. Cross-Layer Multi-Objective Optimization

A full-pipeline grid search evaluated 108 end-to-end parameter configurations across DL, DIP, CN, and Crypto on genuine radiographs. We extracted **26 non-dominated Pareto configurations** balancing ROI SSIM, wire bytes, triage latency, and crypto cost.

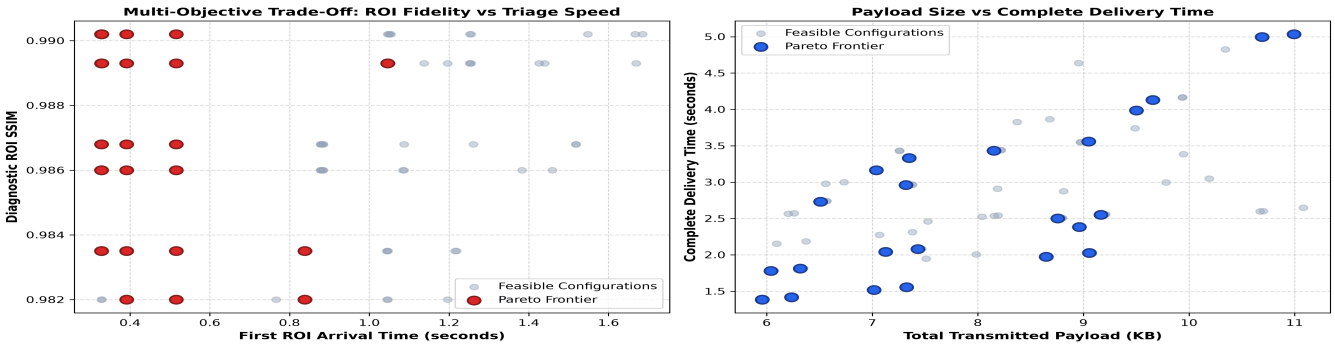


***Figure 2**: Empirical Pareto Frontier (108 configurations): Demonstrates dominant trade-offs between ROI delivery latency, compression quality, and wire bandwidth across priority and FIFO regimes.*

8. Discussion, Limitations & Conclusion

Limitations: (1) The active clinical cohort contains 166 radiographs across 110 patients; prospective clinical multi-center trials will scale to 20,000+ studies. (2) Ascon-AEAD128 is implemented in pure Python for transparency; production deployments will link C/Rust assembly. (3) Standard-cell GE area numbers are pre-synthesis analytical estimates (2.0 GE/XOR). (4) MediLink-DL is an engineering research prototype, not FDA/CE-cleared software.

Conclusion: MediLink-DL successfully demonstrates an end-to-end, reproducible pipeline uniting differentiable DL localization, dual-stream compression, authenticated AEAD ciphers, priority transport, and verified involutory diffusion architectures, providing a robust template for rural telemedicine.

References

[1] World Health Organization, *Telemedicine: Opportunities and Developments in Member States*, Geneva: WHO Press, 2010.

[2] J. P. Cohen, P. Morrison, and L. Dao, "COVID-19 Image Data Collection," *arXiv:2003.11597*, 2020.

[3] R. R. Selvaraju, et al., "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE ICCV*, 2017.

[4] Z. Wang, A. C. Bovik, et al., "Image quality assessment: from error visibility to structural similarity," *IEEE TIP*, vol. 13, no. 4, 2004.

[5] M. Dworkin, "Galois/Counter Mode (GCM) and GMAC," *NIST SP 800-38D*, 2007.

[6] C. Dobraunig, M. Eichlseder, et al., "Ascon v1.2: Lightweight Authenticated Encryption," *NIST SP 800-232*, 2024.

[7] P. Rajpurkar, et al., "CheXNet: Radiologist-level pneumonia detection on chest X-rays," *arXiv:1711.05225*, 2017.

[8] M. Sajadieh, M. Wei, and E. Pasalic, "On construction of involutory MDS matrices," *IEEE TIT*, vol. 58, no. 9, 2012.

[9] C. Beierle, A. Canteaut, and G. Leander, "Lightweight Invertible Diffusion Layers," *IACR ToSC*, vol. 2018, no. 3, 2018.

[10] V. Cauchois and P. Loidreau, "On the number of involutory MDS matrices over finite fields," *Des. Codes Cryptogr.*, 2019.

[11] D. Han, X. Wang, and B. Preneel, "Constructing lightweight MDS and near-MDS matrices," in *Inscrypt 2016*, 2016.